

深度学习在资产定价与组合优化中的应用

一、学习经济结构

将无套利约束、pricing errors、SDF、因子结构或投资者效用嵌入深度学习。

代表文献：

Deep Learning in Asset Pricing

Deep Learning in

Characteristics-Sorted Factor Models

Deep Parametric Portfolio Policies

Deep Learning Statistical Arbitrage

二、学习价格模式

从历史价格、成交量、收益序列或价格图像中学习可预测的趋势和技术分析信号。

代表文献：

Charting by Machines

(Re-)Imag(in)ing Price Trends

三、学习风险情景

用生成式模型学习金融时间序列的分布，尤其关注极端情景和尾部风险。

代表文献：

TAIL-GAN: Learning to Simulate Tail Risk Scenarios

这三类文献的区别在于 金融对象、输入数据、学习目标与损失函数不同。

深度学习服务于不同金融目标

文献类别	代表文献	深度学习学习什么？	与两篇主讲文献的联系
资产定价表示学习	<i>Deep Learning in Asset Pricing</i> <i>Deep Learning in Characteristics-Sorted Factor Models</i>	学习 SDF、非线性因子结构、特征交互、pricing errors，以及宏观状态或公司状态表示。	提供“经济结构 + 深度学习”的基础：模型不只是拟合收益，而是嵌入资产定价约束。
价格图形与趋势预测	<i>Charting by Machines</i> <i>(Re-)Imag(in)ing Price Trends</i>	学习价格图像、历史趋势、技术分析形态中对未来收益有预测力的非线性模式。	与 <i>Deep Learning Statistical Arbitrage</i> 都关注时间序列信号；区别是这里通常直接使用原始价格或收益，而统计套利先构造 factor residuals。
直接组合与交易政策	<i>Deep Parametric Portfolio Policies</i> <i>Deep Learning Statistical Arbitrage</i>	学习组合权重、交易信号、配置规则，并用效用、Sharpe ratio 或均值-方差目标训练。	从 prediction-aware learning 转向 decision-aware learning。
生成式风险建模	<i>TAIL-GAN</i>	学习能够保留 VaR、ES 等尾部风险特征的价格情景分布。	补充风险管理维度：前几类回答如何预测、定价和配置，TAIL-GAN 回答如何生成极端情景并检验策略稳健性。

相同范式，不同金融对象

比较维度	Deep Parametric Portfolio Policies	Deep Learning Statistical Arbitrage
研究对象	投资组合权重：从资产特征和偏好参数直接映射到组合配置	统计套利头寸：从因子残差中提取临时价格偏离并形成交易权重
输入	公司特征，投资者风险厌恶，组合约束	因子模型残差的时间序列，过去 $L = 30$ 日累计残差
深度学习角色	用神经网络扩展 parametric portfolio policies，学习非线性组合规则	用 CNN+Transformer 学习残差中的局部与全局时间序列模式
优化目标	直接最大化投资者效用；风险厌恶内生地约束模型复杂度	直接最大化 Sharpe ratio 或 mean-variance trading objective
风险暴露	通常仍是配置型投资组合，赚取风险溢价与特征溢价	目标是构造 factor-neutral residual portfolios，捕捉相对价格偏离
主要贡献	说明复杂非线性政策何时能优于线性组合政策	统一统计套利为三步：残差组合生成、信号提取、配置决策

两篇主讲文献都不是单纯“预测收益”。它们共同关心的是：如何让深度学习服务于金融决策目标，而不是只服务于统计预测误差。

Deep Learning Statistical Arbitrage

Jorge Guijarro-Ordóñez, Markus Pelger, Greg Zanoliti

Management Science, 2025

Presenter: 赵伟皓

School of Economics and Management, Wuhan University

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Framework

Motivation Statistical arbitrage is widely used, but its profitability and key design elements are hard to evaluate academically.

RQ How much realistic statistical arbitrage exists in U.S. equities, and which model components determine its profitability?

Sub-RQ1 / H1
Residual portfolios
define similar assets

Design
FF / PCA / IPCA residuals

Sub-RQ2 / H2
CNN+Transformer
extracts signals

Design
OU, FFT, CNN+Trans

Sub-RQ3 / H3
OOS profits and
arbitrage alphas

Evidence
Table 1 + Table 2

Contribution A unified economic framework and a deep-learning implementation for tradable statistical arbitrage.

Logic: Residual portfolios define similarity; CNN+Transformer learns time-series deviations; trading-objective allocation turns signals into OOS arbitrage returns.

Why Study Statistical Arbitrage?

Why it matters

- Statistical arbitrage is widely used by hedge funds and investment banks.
- Yet its true profitability is hard to evaluate from public evidence.
- Most successful strategies are proprietary, so their economic sources remain unclear.

Limitations of existing work

- Classical studies often focus on pairs trading or simple mean reversion.
- Portfolio construction, signal extraction, and allocation are usually treated separately.
- Return-prediction models do not directly optimize a trading objective.

Asset-pricing relevance

- The strategy tests temporary deviations from the law of one price.
- It separates residual mispricing from compensation for systematic risk.

Key motivation: This paper studies whether deep learning can build a complete statistical arbitrage system that extracts tradable signals from residual price deviations.

Research Question

Main Research Question

How to systematically construct statistical arbitrage strategies in the US stock market, and which model components truly determine arbitrage returns?

RQ1: How should 'similar assets' be defined and converted into tradable arbitrage portfolios?

RQ2: How should the model extract informative time-series signals from arbitrage portfolios?

RQ3: Will the resulting strategies earn economically large and statistically significant profits?

Organizing principle: Each sub-question maps directly into one model component.

Contributions

Related literature	What prior studies do	How this paper extends it
Classical statistical arbitrage: pairs trading, PCA residual trading, OU processes; Yeo and Papanicolaou (2017)	Study portfolio generation, signal extraction, and allocation mostly as separate problems	Builds a unified framework and defines arbitrage portfolios as residual portfolios from asset pricing models
Machine learning in asset pricing and return prediction; Chen et al. (2020), Gu et al. (2020)	Estimate risk premia or predict returns using characteristics and flexible models	Focuses on residual time-series deviations around risk premia, rather than cross-sectional expected returns
Deep learning for time-series modeling; Vaswani et al. (2017)	CNNs and Transformers learn local and global patterns in sequential data	Adapts CNN+Transformer to residual time series and optimizes the model under an economic trading objective

Marginal contribution: The paper contributes not only a new architecture, but also an economic decomposition that makes statistical arbitrage methods comparable.

Hypotheses

Research question	Testable hypothesis
RQ1: Define similar assets	H1: Residual portfolios from sufficiently rich factor models provide better arbitrage inputs than raw stock returns.
RQ2: Extract signals	H2: A CNN+Transformer signal extractor captures residual time-series patterns better than parametric or pre-specified-filter models.
RQ3: Test profits	H3: The resulting strategy earns large out-of-sample returns and significant alphas not explained by conventional risk factors.

Logic: The hypotheses follow the model pipeline: residual construction → signal extraction → tradable arbitrage profits.

Three Model Components

1. Arbitrage portfolios

$$\varepsilon_t = \Phi_{t-1} R_t$$

- Use factor residuals.
- Define similarity economically.
- Trade deviations from mimicking portfolios.

2. Signal extraction

$$\theta_{n,t-1} = u(\varepsilon_{n,t-1}^L)$$

- Use lagged residual history.
- Learn time-series patterns.
- Summarize trading-relevant information.

3. Allocation decision

$$w_{n,t-1}^\varepsilon = w^\varepsilon(\theta_{n,t-1})$$

- Map signals to weights.
- Optimize Sharpe or mean-variance objective.
- Incorporate constraints and costs.

Stock returns → Factor residual portfolios → Time-series signals → Allocation weights → Trading objectives

From Similar Assets to Residual Portfolios

Economic definition of similarity

- Similar assets have similar exposure to systematic risk factors.
- A target stock is compared with a factor-mimicking portfolio.
- Residuals are tradable because factors are represented by traded or factor-mimicking portfolios.

Factor models used to construct residuals

- **Fama–French factors:** observed traded factors.
- **PCA factors:** latent unconditional factors from return comovement.
- **IPCA factors:** conditional latent factors using firm characteristics.

$$R_{n,t} = \beta'_{n,t-1} F_t + \varepsilon_{n,t}, \quad \varepsilon_t = \Phi_{t-1} R_t$$

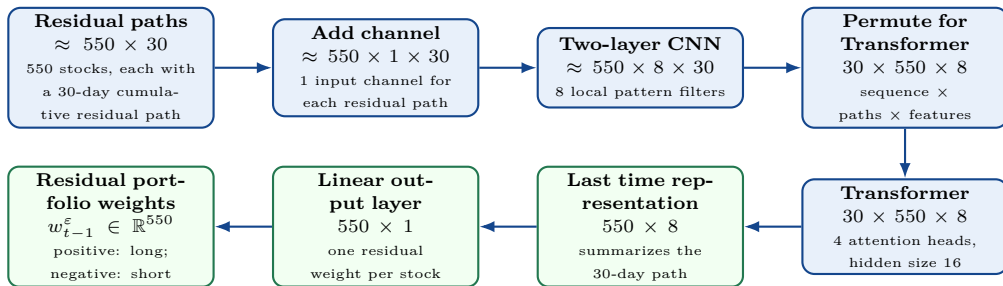
Residual portfolios convert the idea of “similar assets” into a tradable object suitable for time-series arbitrage.

Signal and Allocation Models

Model	Signal extraction	Allocation policy
OU + Threshold	Parametric OU mean-reversion signal	Fixed threshold rule
Fourier + FFN	Pre-specified frequency decomposition of residual history	Feedforward neural network
CNN + Transformer	Data-driven local filters and global attention patterns	Feedforward neural network under trading objective
CNN component: learns local trend and reversal filters from short residual windows.		Transformer component: combines local filters into global dependency patterns through attention heads.

Key design choice: The main innovation is learning trading-relevant time-series signals, not simply adding a neural network to a classical allocation rule.

CNN+Transformer Pipeline



Interpretation: CNN extracts local trend and reversal features from each 30-day residual path. The Transformer uses attention to combine these local features into a global trading signal. The final layer maps the signal into residual portfolio weights.

Data and Evaluation Protocol

Data and sample

- Raw CRSP data and characteristics begin before 1998 for factor estimation.
- Residuals generated: 1998–2016.
- OOS trading evaluation: 2002–2016.
- Universe: roughly 550 largest and most liquid U.S. stocks.
- Addit. inputs: 46 firm characteristics.

Training and evaluation

- Daily trading at market close.
- Signal lookback window: $L = 30$ days.
- Deep models trained on rolling 1,000-day windows.
- Re-estimation every 125 days.
- Metrics: annualized Sharpe ratio, mean return, volatility, alpha.

H1: Residual Portfolios Improve Arbitrage Inputs

Test of H1: Residual portfolios from sufficiently rich factor models provide better arbitrage inputs than raw stock returns.

Model	Factors	Fama-French			PCA			IPCA		
	K	SR	μ	σ	SR	μ	σ	SR	μ	σ
CNN + Trans	0	1.64	13.7%	8.4%	1.64	13.7%	8.4%	1.64	13.7%	8.4%
	1	3.68	7.2%	2.0%	2.74	15.2%	5.5%	3.22	8.7%	2.7%
	3	3.13	5.5%	1.8%	3.56	16.0%	4.5%	3.93	8.6%	2.2%
	5	3.21	4.6%	1.4%	3.36	14.3%	4.2%	4.16	8.7%	2.1%
	8	2.49	3.4%	1.4%	3.02	12.2%	4.0%	3.95	8.2%	2.1%
	10	—	—	—	2.81	10.7%	3.8%	3.97	8.0%	2.0%
	15	—	—	—	2.30	7.6%	3.3%	4.17	8.4%	2.0%

Main takeaway: Residualization removes systematic components and creates time-series objects with more learnable arbitrage patterns than raw stock returns.

H2: CNN+Trans Dominates Benchmark Models

Test of H2: CNN+Transformer captures residual time-series patterns better than parametric or pre-specified-filter models.

Model	Factors K	Fama-French			SR	PCA			SR	IPCA		
		SR	μ	σ		SR	μ	σ		SR	μ	σ
CNN + Trans	5	3.21	4.6%	1.4%		3.36	14.3%	4.2%		4.16	8.7%	2.1%
	8	2.49	3.4%	1.4%		3.02	12.2%	4.0%		3.95	8.2%	2.1%
	10	–	–	–		2.81	10.7%	3.8%		3.97	8.0%	2.0%
Fourier + FFN	5	1.66	3.1%	1.8%		1.98	12.4%	6.3%		1.90	7.7%	4.1%
	8	1.90	3.1%	1.6%		1.95	10.1%	5.2%		1.94	7.8%	4.0%
	10	–	–	–		1.71	8.2%	4.8%		1.93	7.6%	3.9%
OU + Thresh	5	0.38	0.9%	2.3%		0.73	4.4%	6.1%		0.97	3.8%	4.0%
	8	1.16	2.8%	2.4%		0.87	4.4%	5.1%		0.91	3.5%	3.8%
	10	–	–	–		0.63	2.9%	4.6%		0.86	3.1%	3.6%

Main takeaway: The gain comes mainly from flexible time-series signal extraction. A flexible allocation rule alone cannot compensate for a weak signal.

H3: Arbitrage Profits Are Not Standard Factor Premia

	Fama-French		PCA		IPCA	
	K=0	K=5	K=0	K=5	K=0	K=5
α (%)	11.6	4.5	11.6	14.1	11.6	8.3
μ (%)	13.7	4.6	13.7	14.3	13.7	8.7
t_α	6.4***	12***	6.4***	13***	6.4***	16***
t_μ	6.3***	12***	6.3***	13***	6.3***	16***
R^2 (%)	30.3	2.3	30.3	1.3	30.3	3.9

Table 2 extract: CNN+Transformer alphas relative to Fama–French 8-factor model.

Main takeaway: The strategy earns conditional arbitrage alphas from residual time-series patterns, rather than repackaging standard factor premia.¹

¹The conclusion also survives trading frictions: with IPCA-5 residuals, the cost-adjusted CNN+Transformer strategy still achieves an annual Sharpe ratio around 1.11.

OOS Performance of CNN+Trans with Trading Frictions

K	IPCA factor model					
	Sharpe ratio			Mean-variance		
	SR	μ	σ	SR	μ	σ
0	0.52	8.5%	16.3%	0.22	2.6%	11.9%
1	0.85	5.9%	6.9%	0.86	5.5%	6.4%
3	1.24	6.6%	5.4%	1.16	6.9%	5.9%
5	1.11	5.5%	5.0%	1.02	5.3%	5.3%
10	0.98	5.1%	5.2%	1.04	5.4%	5.2%
15	0.94	4.8%	5.1%	1.02	5.1%	5.0%

This table shows the out-of-sample annualized Sharpe ratio (SR), mean return (μ), and volatility (σ) for the CNN+Transformer model with trading frictions on IPCA residuals. The models use either the Sharpe ratio or mean-variance objective function with trading costs

$$\text{cost}(w_{t-1}^R, w_{t-2}^R) = 0.0005 \|w_{t-1}^R - w_{t-2}^R\|_{L^1} + 0.0001 \|\min(w_{t-1}^R, 0)\|_{L^1}.$$

为什么这篇文章能发到 *MS*?

1. **重要性**: 统计套利是业界真实使用的高价值策略, 文章研究的是一个有经济意义和实践价值的决策问题。
2. **框架贡献**: 文章不是单纯提出一个深度学习模型, 而是把统计套利拆成组合构造、信号提取和交易配置三个模块。
3. **方法直接服务经济结构**: 用资产定价残差定义相似资产, 再学习临时偏离的时间序列模式。
4. **模型可解释性**: CNN filters 和 Transformer attention heads 可以对应到趋势、反转和非对称交易反应。

总结: 这篇文章的亮点不只是“深度学习表现好”, 而是它把机器学习嵌入了一个清晰的金融经济学问题。

拓展

1. **中国市场复现与制度约束：** A 股市场存在卖空限制、涨跌停、流动性分层和交易成本差异，统计套利的可交易性本身值得研究。
2. **因子构造与交易目标联合优化：** 原文先构造 residual portfolios，再学习信号和交易策略。未来可以让 residual construction 也服务于 trading objective。
3. **引入中国市场特色数据：** 可以使用行业链关系、公告文本、分析师报告、融资融券、订单流或高频流动性变量改进 residual construction 和 signal extraction。

潜在论文方向： 在中国市场约束下，检验深度学习统计套利是否仍能产生可交易的条件 alpha。

中国市场复现

中国 A 股样本

- 股票池：A 股月度市值排序 top 550
- 样本区间：2000–2024 年日度数据
- 当前样本外区间：约 2022–2024 年
- 收益频率：日度超额收益
- 数据来源：CSMAR 日个股收益、无风险利率、五因子数据
- 交易约束：暂不加入交易成本、卖空约束、涨跌停限制

与原文设定的对应关系

- 原文 Table 1 → 中国市场日频残差组合表现。
- 原文 Table 2 → 中国市场因子 alpha 检验。
- Fama–French 残差 → 中国五因子 FF-like 残差。
- PCA 残差 → 日滚动 PCA + 60 日 loading 回归。
- IPCA 残差 → 46 个中国公司特征的 IPCA-like 残差。
- 训练设定：1000 日训练窗口，125 日重训，lookback = 30。

中国市场复现 Table 1

原文：Table 1 对应结果

Residual	K	OU	Fourier	CNN
Raw return	0	-0.18	0.36	1.64
PCA	1	0.21	0.80	2.74
PCA	3	0.77	1.66	3.56
PCA	5	0.73	1.98	3.36
PCA	8	0.87	1.95	3.02
PCA	10	0.63	1.71	2.81
PCA	15	0.62	1.14	2.30
Fama–French	5	0.38	1.66	3.21
IPCA	5	0.97	1.90	4.16

中国市场：目前已完成结果

Residual	K	OU	Fourier	CNN
Raw return	0	-0.97	-0.34	0.08
PCA	1	-1.20	1.85	-0.28
PCA	3	0.06	-0.15	-0.23
PCA	5	-0.16	0.66	0.21
PCA	8	1.17	1.65	0.96
PCA	10	1.16	2.50	1.19
PCA	15	1.03	2.93	1.00
China FF-like	5	-2.12	3.90	2.93
IPCA46-like	5	-1.02	0.14	0.46

美国市场；Table 1；年化 Sharpe ratio。

top 550 A 股；表中为年化 Sharpe ratio。

原文中 CNN+Transformer 在 FF、PCA、IPCA 残差上均明显领先；中国市场复现中，残差套利信号主要出现在 China FF-like 五因子残差和高维 PCA 残差上，但目前最强的是 Fourier+FFN，而不是 CNN+Transformer。

中国市场复现：最佳结果对比

原文：代表性最佳结果

Model	K	SR	μ	σ
CNN+Trans	5	4.16	8.7%	2.1%
Fourier+FFN	5	1.90	7.7%	4.1%
OU+Thresh	5	0.97	3.8%	4.0%

中国市场：当前各模型最佳结果

Model	Residual	SR	μ	σ
CNN+Trans	FF5	2.93	11.3%	3.9%
Fourier+FFN	FF5	3.90	15.0%	3.8%
OU+Thresh	PCA8	1.17	4.9%	4.1%

IPCA-5 残差；Sharpe ratio 目标函数。

从目前已完成的中国市场复现结果中，按各模型最高 SR 选取。

中国市场下最好的结果在量级上已经接近原文基准模型，尤其 Fourier+FFN 的 SR 达到 3.90；但模型排序发生变化，CNN 并未成为当前最优。这说明残差套利信号存在，但中国市场中的有效时间序列结构、残差构造和原文美国市场并不完全一致。

残差组合

核心直觉

- 为每只股票构造一个“参照组合”。²
- 参照组合复制该股票的系统性风险暴露。
- 残差组合衡量目标股票相对参照组合的偏离。

$$R_{n,t} = \beta_{n,t-1}^\top F_t + \varepsilon_{n,t}$$

$$F_t = (w_{t-1}^F)^\top R_t$$

$$\varepsilon_t = R_t - \beta_{t-1} (w_{t-1}^F)^\top R_t = \Phi_{t-1} R_t$$

示例

- 股票 A 今日收益: $R_{A,t} = 2.0\%$ 。
- 因子模型给出的基准收益:

$$\beta_A^\top F_t = 1.6\%.$$

- 残差收益:

$$\varepsilon_{A,t} = 2.0\% - 1.6\% = 0.4\%.$$

- 含义: 股票 A 相对其参照组合多涨了 0.4%。

Residual portfolio 是一个可交易的 long-short 组合: 做多目标股票, 做空其 mimicking portfolio。统计套利关注的是这种相对偏离是否会短期回归。

²FF 模型包括 K=1,3,5,8: 市场因子、FF3、FF5, 以及加入 Mom、ST-rev、LT-rev 后的 FF8
PCA 使用 K=1,3,5,8,10,15。用过去 252 天估计相关矩阵并提取因子, 股票 loading 用过去 60 天日收益滚动回归估计
IPCA 的 loading 是 t-1 期 46 个公司特征的线性函数; 每年用过去 240 个月的月度收益和特征重新估计 IPCA。

OU、FFT 与 CNN+Transformer 如何处理残差路径

- 共同输入：每只股票过去 $L = 30$ 天的累计残差路径

$$x_{n,t-1} = (x_{n,t-30}, \dots, x_{n,t-1}).$$

单日约有 550 只股票，因此输入约为 550×30 。

- OU**：将 30 天路径压缩成均值回复特征

$$x_{1:30} \rightarrow (X_L, \mu, \sigma, R^2).$$

例：若 $X_L = -0.80\%$ 、 $\mu = -0.10\%$ 、 $\sigma = 0.35\%$ 、 $R^2 = 0.60$ ，则当前残差明显低于均值，模型倾向判断其可能回升。

- FFT**：将 30 天路径分解为不同频率成分

$$x_{1:30} \rightarrow (a_0, \dots, a_{15}, b_1, \dots, b_{14}).$$

例：若低频系数较强，说明残差路径主要包含慢周期趋势或反转；若高频系数较强，说明路径更像短期震荡。

- CNN+Transformer**：直接学习路径中的局部形态和全局依赖

$$550 \times 30 \rightarrow 550 \times 8 \times 30 \rightarrow 550.$$

CNN 识别局部上涨、下跌和反转；Transformer 判断这些局部模式在 30 天窗口中如何组合。

- 核心区别**：OU 预设“均值回复”；FFT 预设“频率分解”；CNN+Transformer 从数据中学习可交易模式。

OU 参数化模型

经典均值回复交易策略 (Avellaneda and Lee (2010)):

残差建模: 每个残差被建模为 Ornstein-Uhlenbeck (OU) 过程:

$$dX_t = \kappa(\mu - X_t)dt + \sigma dB_t$$

交易信号与仓位分配

分配权重基于标准化比率 $\frac{X_t - \mu}{\sigma/\sqrt{2\kappa}}$ 的阈值规则:

$$w^\epsilon(\theta) = \begin{cases} +1, & \text{if } \frac{X_t - \mu}{\sigma/\sqrt{2\kappa}} > c_{\text{thres}} \\ -1, & \text{if } \frac{X_t - \mu}{\sigma/\sqrt{2\kappa}} < -c_{\text{thres}} \\ 0, & \text{otherwise} \end{cases}$$

其中阈值参数 c_{thres} 通过最优化选取。

模型局限

- 参数化模型可能存在误设风险 (例如, 数据中存在趋势项、多频率均值回复等复杂模式);
- 分配函数形式过于 restrictive (受限), 难以刻画非线性动态。

FFT + FFN

信号提取：通用时间序列模型

- 采用预指定的线性滤波器提取信号：

$$\theta_l = \sum_{j=1}^L W_j^{\text{filter}} x_j, \quad W_j^{\text{filter}} \in \mathbb{R}^{L \times L}$$

该框架可涵盖 ARMA 模型、离散化 OU 过程等传统时间序列模型。

- 针对均值回复模式，采用频率滤波器（快速傅里叶变换，FFT）进行频率分解：

$$x_l = a_0 + \sum_{j=1}^{L/2-1} \left(a_j \cos\left(\frac{2\pi j}{L} l\right) + b_j \sin\left(\frac{2\pi j}{L} l\right) \right) + a_{L/2} \cos(\pi l)$$

- 提取的信号即为长、短期反转模式的“载荷”：

$$\theta^{\text{FFT}}(x) = (a_0, \dots, a_{L/2}, b_1, \dots, b_{L/2-1})$$

仓位分配：灵活非参数函数

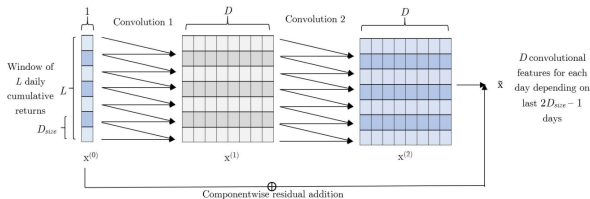
分配函数 w^c 采用带有正则化的灵活非参数形式，通过前馈神经网络（FFN）进行估计：

$$w^{c|\text{FFT}}(\theta^{\text{FFT}}) = g^{\text{FFN}}(\theta^{\text{FFT}})$$

模型局限

预指定滤波器的选择限制了模型能够捕捉的时间序列模式，可能无法充分挖掘数据中的复杂非线性动态。

卷积神经网络 (CNN)

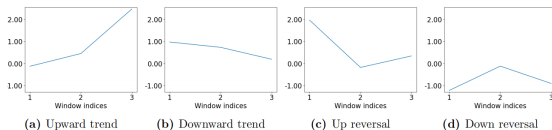


- 将 CNN 应用于残差时间序列 ϵ_t^L 。

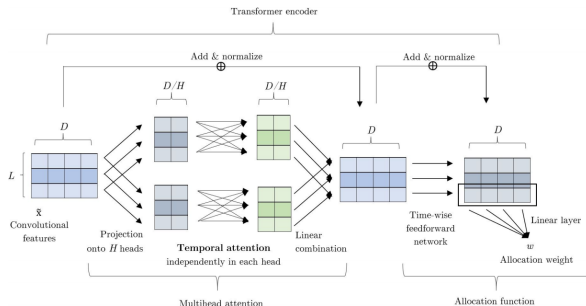
时间序列的局部模式表示经过 CNN 处理后，原始序列 x 被重新表达为 D 个局部模式的组合：

$$\tilde{x} \in \mathbb{R}^{L \times D}$$

每一列对应一个局部模式，可捕捉不同时间尺度下的形态特征（如局部趋势、曲率等）。



Transformer Network



捕获全局时序依赖 Transformer 的核心作用是捕获局部模式之间的时序依赖关系，从而从局部特征中提取全局结构。注意力机制与全局模式给定 CNN 提取的局部模式表示 $\tilde{x} \in \mathbb{R}^{L \times D}$ ，Transformer 通过注意力机制计算全局模式：

$$h_i = \sum_{l=1}^L \alpha_{i,l} \tilde{x}_l$$

其中注意力权重定义为： $\alpha_{i,l} = \alpha_i(\tilde{x}_L, \tilde{x}_l)$ ， $l = 1, \dots, L$ ， $i = 1, \dots, H$ 。

注意力头与模式因子

- H 个全局模式由“注意力权重” $\alpha_i \in \mathbb{R}^L$ 来刻画和指定。
- 每个注意力头 h_i 可视之为对某一特定“模式因子” α_i 的“载荷” (loadings)，反映了该全局模式在输入序列中的激活强度。